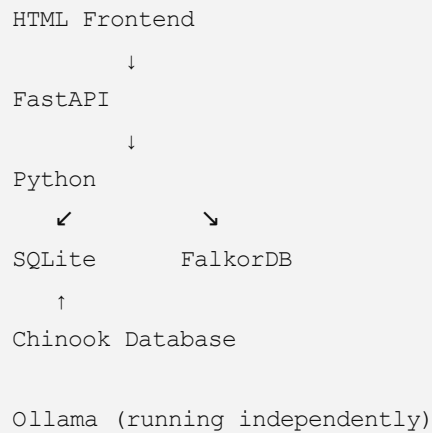


Excellent. Let's make this completely **hands-on**.

Our goal for **Phase 1** is **not** to build an agent yet.

At the end of this phase, you should have this running locally:



Success criteria:

- ☒ Ollama installed
- ☒ Local LLM responding
- ☒ Docker installed
- ☒ FalkorDB running
- ☒ Chinook database downloaded
- ☒ FastAPI application running
- ☒ HTML page communicating with FastAPI
- ☒ SQLite queries working
- ☒ FalkorDB queries working

Step 0: Verify your machine

Before we install anything, please tell me:

CPU

Bash

```
wmic cpu get name
```

RAM

Bash

```
systeminfo | findstr "Total Physical Memory"
```

GPU

Bash

```
wmic path win32_VideoController get name
```

While you're checking that, let's start preparing the environment.

Step 1: Install Python

Verify Python:

Bash

```
python --version
```

Expected:

```
Python 3.11.x
```

If Python isn't installed, install **Python 3.11**.

Verify pip:

Bash

```
pip --version
```

Step 2: Install Git

Verify:

Bash

```
git --version
```

Expected:

```
git version 2.x.x
```

Step 3: Install VS Code

Verify:

Bash

```
code --version
```

Install these VS Code extensions:

```
Python  
Pylance  
Docker  
SQLite Viewer  
HTML CSS Support  
GitLens
```

Step 4: Install Ollama

Verify:

Bash

```
ollama --version
```

Expected:

```
ollama version x.x.x
```

Pull a model

For your AI architect learning project, I recommend:

RAM	Model
16 GB	qwen3:4b
32 GB	qwen3:8b
64 GB	qwen3:14b

Pull the model:

Bash

```
ollama pull qwen3:8b
```

Test Ollama

Run:

Bash

```
ollama run qwen3:8b
```

Ask:

```
Write a SQL query to retrieve the top 5 customers by revenue.
```

If you receive a response, Ollama is working.

Exit:

```
/bye
```

Step 5: Install Docker Desktop

Verify:

Bash

```
docker --version
```

Expected:

```
Docker version x.x.x
```

Verify Docker is running:

Bash

```
docker ps
```

Expected:

```
CONTAINER ID
```

An empty list is fine.

Step 6: Start FalkorDB

Pull the image:

Bash

```
docker pull falkordb/falkordb:latest
```

Start the container:

Bash

```
docker run \  
-p 6379:6379 \  
--name falkordb \  
falkordb/falkordb:latest
```

Open another terminal.

Verify:

Bash

```
docker ps
```

Expected:

```
CONTAINER ID  
falkordb
```

Step 7: Create the project

Create the folder:

Bash

```
mkdir agentic-text2sql  
cd agentic-text2sql
```

Create the virtual environment:

Bash

```
python -m venv .venv
```

Activate it:

Bash

```
.venv\Scripts\activate
```

You should see:

```
(.venv)
```

Step 8: Create the folder structure

```
agentic-text2sql  
|  
├─ app  
|  
├─ data  
|  
├─ frontend  
|  
├─ tests  
|  
├─ .env  
|  
├─ requirements.txt  
|  
└─ README.md
```

Create them:

Bash

```
mkdir app
mkdir data
mkdir frontend
mkdir tests

type nul > .env
type nul > requirements.txt
type nul > README.md
```

Step 9: Install dependencies

Install everything at once:

Bash

```
pip install fastapi
pip install uvicorn
pip install sqlalchemy
pip install ollama
pip install falkordb
pip install python-dotenv
pip install pydantic
pip install pandas
pip install pytest
```

Save them:

Bash

```
pip freeze > requirements.txt
```

Step 10: Download Chinook

Create:

```
agent-ic-text2sql
├── data
│   └── chinook.db
```

Download the SQLite database from:

[Chinook database releases](#)

Place **Chinook_Sqlite.sqlite** inside the **data** folder and rename it:

```
chinook.db
```

Verify the file exists:

Bash

```
dir data
```

Expected:

```
chinook.db
```

Step 11: Test SQLite

Create:

```
app/test_sqlite.py
```

Add:

Python

Run

```
import sqlite3

connection = sqlite3.connect("../data/chinook.db")

cursor = connection.cursor()

query = """
SELECT Name
FROM Artist
LIMIT 10
"""

cursor.execute(query)

rows = cursor.fetchall()

for row in rows:
    print(row)

connection.close()
```

Run:

Bash

```
cd app
```

```
python test_sqlite.py
```

Expected:

```
AC/DC
Accept
Aerosmith
Alanis Morissette
Alice In Chains
...
```

Step 12: Test FalkorDB

Create:

```
app/test_falkordb.py
```

Add:

Python

Run

```
from falkordb import FalkorDB

db = FalkorDB(
    host="localhost",
    port=6379
)

graph = db.select_graph("text2sql")

graph.query("""
CREATE (:Table {
    name: 'Artist',
    description: 'Contains artist information'
})
""")

result = graph.query("""
MATCH (t:Table)
RETURN t.name
""")
```

```
for row in result.result_set:
```

Run:

Bash

```
python test_falkordb.py
```

Expected:

```
Artist
```

Step 13: Create FastAPI

Create:

```
app/main.py
```

Add:

Python

Run

```
from fastapi import FastAPI

app = FastAPI()

@app.get("/")
def home():

    return {
        "message": "Agentic Text-to-SQL API"
    }
```

Run:

Bash

```
uvicorn main:app --reload
```

Open:

```
http://127.0.0.1:8000
```

Expected:

JSON

```
{  
  "message": "Agentic Text-to-SQL API"  
}
```

Step 14: Create the HTML frontend

Create:

```
frontend/index.html
```

Add:

HTML

```
<!DOCTYPE html>  
<html>  
  <head>  
    <title>Text-to-SQL</title>  
  </head>  
  
  <body>  
  
    <h1>Agentic Text-to-SQL</h1>  
  
    <input  
      type="text"  
      id="question"  
      placeholder="Ask a question"  
    />  
  
    <button onclick="sendQuestion()" ">  
      Submit
```

Open it in the browser.

Expected:

```
Agentic Text-to-SQL  
  
[ Ask a question ]  
  
[ Submit ]
```

Phase 1 checklist

- ☐ Python installed
- ☐ Git installed
- ☐ VS Code installed
- ☐ Ollama installed
- ☐ Qwen model downloaded
- ☐ Docker installed
- ☐ FalkorDB running
- ☐ Virtual environment created
- ☐ Dependencies installed
- ☐ Chinook database downloaded
- ☐ SQLite tested
- ☐ FalkorDB tested
- ☐ FastAPI tested
- ☐ HTML frontend created

Please stop after Step 4 and tell me:

1. CPU
2. RAM
3. GPU
4. Whether Ollama was installed successfully
5. Whether `qwen3:8b` runs successfully

Those answers will determine which model we should use before moving to the remaining steps.